# A WEAKLY ADIC RING WHOSE IDEAL OF DEFINITION HAS FINITE DIMENSIONAL SQUARE QUOTIENT IS ADIC

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof displays one containment; it lies inside the counted span and is reproduced verbatim.

*Editorial note.* The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- The characterisation of weakly adic rings, in particular the role of an ideal of definition with open power closures. Same chapter, Lemma lemma-weakly-adic, invoked at proof lines 1720-1721. Supplies the framework in which the induction is carried out.
- A weakly adic ring whose ideal of definition is finitely generated is adic. Same chapter, Lemma lemma-weakly-adic-finite-generation-bis, invoked at proof lines 1733-1734. Gives the conclusion once finite generation of an ideal of definition has been established.
- The square of a sum of ideals, and the closure of a sum. Standard, used at proof lines 1724-1728. Underlies the induction which shows that the ideal generated by the chosen elements has the given ideal as its closure.

*Source reference.* Original lemma 0GXF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1633–1633. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0GXG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1699–1699. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $A$ be a weakly adic topological ring. Let $I$ be an ideal of definition such that $I/I_2$ is a finitely generated module where $I_2$ is the closure of $I^2$. Then $A$ is adic and has a finitely generated ideal of definition.*

*Date*: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* We use the characterization of Lemma 0GXF without further mention. Choose $f_1, \ldots, f_r \in I$ which map to generators of $I/I_2$. Set $I' = (f_1, \ldots, f_r)$. We have $I' + I_2 = I$. Then $I_2$ is the closure of $I^2 = (I' + I_2)^2 \subset I' + I_3$ where $I_3$ is the closure of $I^3$. Hence $I' + I_3 = I$. Continuing in this fashion we see that $I' + I_n = I$ for all $n \geq 2$ where $I_n$ is the closure of $I^n$. In other words, the closure of $I'$ in $A$ is $I$. Hence the closure of $(I')^n$ is $I_n$. Thus the closures of $(I')^n$ are a fundamental system of open ideals of $A$. We conclude by Lemma 0GXG. $\square$